

EE 372 Assignment 1: Estimation Variance of the Sample Mean

Student Name: Ahmet Asim Sinan Student ID: 2023401216

1 Introduction

In this assignment, we examine the behavior of the sample mean (M_n) as an estimator. The simulations are implemented using the Python programming language, by utilizing scientific computing libraries such as `numpy` for numerical operations, `matplotlib.pyplot` for data visualization, and `scipy.stats` for distribution modeling and statistical analysis [2]. By utilizing Monte Carlo simulations with various sample sizes ($n \in \{10, \dots, 100\}$) and repetition counts (N), we verify the theoretical properties of the estimator across Gaussian, Exponential, and Uniform distributions, all configured with $E[X] = 2$ and $Var[X] = 4$.

2 Part A: Analysis of Estimation Variance

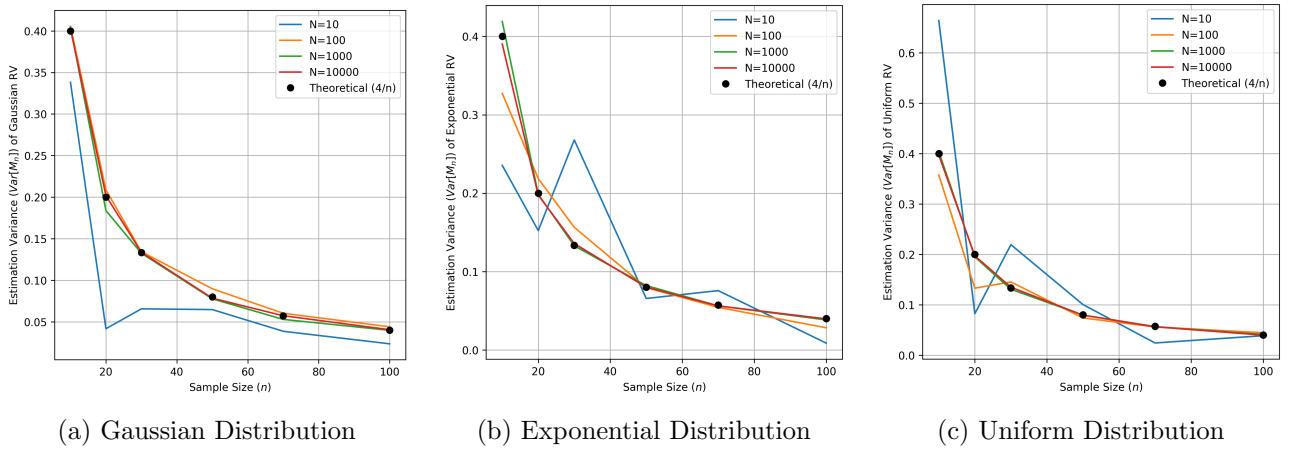


Figure 1: Empirical estimation variance vs. theoretical $Var[M_n] = 4/n$ for various distributions.

2.1 Discussion on Variance Behavior (n and N)

The results illustrate two distinct types of convergence essential to statistical estimation. First, the estimation variance decreases as the sample size n increases. Look for the theoretical $Var[X]/n$ curve. This demonstrates that M_n is a consistent estimator, providing an empirical validation of the Weak Law of Large Numbers (WLLN).

Second, the reliability of the Monte Carlo simulation is governed by the repetition count N . While $N = 10$ results show significant statistical fluctuations (noise) due to the small number of realized paths, $N = 10,000$ yields empirical variances that align almost perfectly with the theoretical expectation. As noted in the core text: “Although the result of a single experiment is unpredictable, predictable patterns emerge as we collect more and more data” [1]. This reinforces that while the sample mean is a random variable, the more information that we can derive from it is highly predictable over many repetitions.

3 Part B: Gaussian Convergence and Histograms

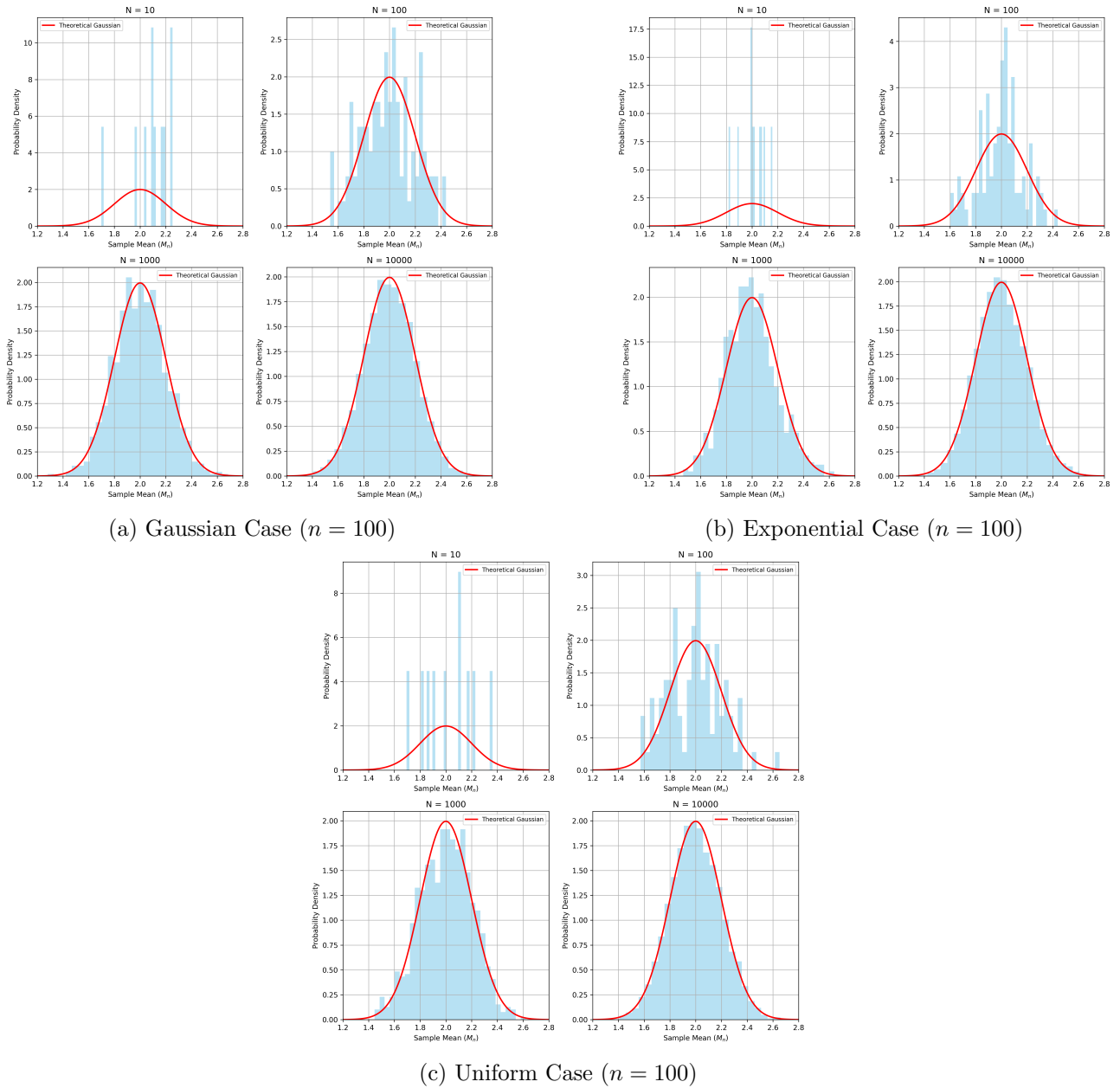


Figure 2: Sample Mean Histograms ($n = 100$) vs. Theoretical Gaussian PDF ($\mu = 2, \sigma = 0.2$).

3.1 Analysis of Gaussian Convergence (CLT)

The histograms reveal the most powerful theorem in our previous course EE 214: the Central Limit Theorem (CLT). Regardless of the original distribution's shape, the distribution of the sample mean M_n converges to a Gaussian shape as the sample size n increases [1]. For the $n = 100$ case, the theoretical standard error is $\sigma_{M_n} = \sqrt{4/100} = 0.2$.

While the theoretical approximation is driven by n , our simulations demonstrate that increasing the repetition count N allows the empirical histogram to more accurately reflect this underlying bell-shaped density, effectively filtering out statistical noise. The symmetry and "bell-shape" observed in the histograms for all three cases provide clear empirical proof that the sum of independent and identically distributed (i.i.d.) variables converges to a normal distribution [1, 3].

4 Confidence Interval Verification

Utilizing the Gaussian approximation for $n = 100$, we construct a 95% confidence interval. For $\alpha = 0.05$, the critical value $z_{\alpha/2} = 1.96$ is used, yielding the interval $\mu \pm 1.96\sigma_{M_n} = [1.608, 2.392]$. The empirical results for $N = 1000$ repetitions are summarized in Table 1.

Distribution	Empirical % ($N = 10^3$)	Target ($1 - \alpha$)	Chebyshev Bound
Gaussian	95.70%	95%	$\geq 73.97\%$
Exponential	95.10%	95%	$\geq 73.97\%$
Uniform	95.30%	95%	$\geq 73.97\%$

Table 1: Confidence Interval verification for $n = 100, N = 1000$. The results demonstrate that the CLT-based interval is far more precise than the conservative Chebyshev inequality.

5 Conclusion

This study confirms that M_n is an unbiased and consistent estimator of the population mean. It is unbiased because its expected value equals the true mean ($E[M_n] = \mu$), and consistent because its estimation variance $Var[M_n]$ approaches zero as $n \rightarrow \infty$ [3]. We observed that the distribution of the estimate converges to Gaussian due to the CLT, a phenomenon that becomes visually evident in our histograms as N increases. While the Chebyshev inequality provides a distribution-independent bound, the Gaussian model allows for much tighter and more efficient confidence intervals, which is crucial for practical engineering applications.

References

- [1] Yates, R. D., & Goodman, D. J. *Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers*. Wiley.
- [2] Kenett, R. S., Zacks, S., & Gedeck, P. *Modern Statistics: A Computer-Based Approach with Python*. Springer.
- [3] Saraçlar, M. *EE 372 Random Signals: Parameter Estimation Using the Sample Mean* [Lecture Slides]. Bogazici University.
- [4] Gemini (Generative AI). Assisted in Python syntax knowledge and LaTeX report structuring.